Secure Microfinance and Welfare Management Platform Using Blockchain and AI

R26-IT-017

Project Proposal Report

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May 2026

1.DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

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Dr.Darshana Kasthurirathna	13th March 2026	

2.ABSTRACT (Maximum 300 words)

Microfinance and welfare management systems play a critical role in supporting economically vulnerable communities. However, many existing systems rely on centralized databases and manual verification processes, which often lack transparency, security, and data integrity. In such systems, administrative decisions related to loan approvals and eligibility verification may be vulnerable to unauthorized modification, data manipulation, or record inconsistencies. These limitations can reduce trust in the system and create challenges in maintaining reliable audit trails.

This research proposes a Blockchain-Based Distributed Data Layer as a component of the SmartGrama platform, which aims to enhance transparency and security in microfinance and welfare management processes. The proposed system utilizes a permissioned blockchain network implemented using Hyperledger Fabric to record verification decisions and loan eligibility outcomes as immutable blockchain transactions. By storing critical verification records in a distributed ledger, the system ensures that once data is recorded, it cannot be modified or deleted, thereby improving trust and accountability among stakeholders.

To address privacy concerns, the system adopts a hybrid storage architecture that separates sensitive personal information from blockchain records. Personal data such as resident name, national identification number (NIC), address, and supporting documents are stored in an off-chain database, while cryptographic hashes and verification metadata are recorded on the blockchain. This approach ensures both data privacy and transparency.

The proposed solution improves the reliability of administrative verification processes and provides a tamper-proof audit trail for welfare and microfinance decisions. By integrating blockchain technology with a hybrid storage model, this research contributes to the development of secure and transparent digital governance systems suitable for public sector financial services.

3.ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor Dr. Darashana Kasthurirathna for his valuable guidance, continuous support, and constructive feedback throughout the development of this research proposal. His expert knowledge, encouragement, and insightful suggestions greatly helped in shaping the direction and quality of this study.

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My sincere thanks also go to my group members and colleagues for their encouragement, cooperation, and valuable discussions throughout the research process.

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7.LIST OF ABBREVIATIONS

Abbreviation	**Meaning**
AI	Artificial Intelligence
API	Application Programming Interface
DLT	Distributed Ledger Technology
GN	Grama Niladhari
ML	Machine Learning
NLP	Natural Language Processing
P2P	Peer-to-Peer
SDG	Sustainable Development Goals
SHA	Secure Hash Algorithm

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1.INTRODUCTION

1.1 Background and Context

Modern inform ation systems across a variety of sectors, including supply chain management, finance, governance, and healthcare, have been profoundly altered by the quick growth of digital technologies. The majority of conventional systems rely on centralized databases, where system data is managed and controlled by a single authority. Despite their widespread use, centralized systems frequently encounter a number of difficulties, including security flaws, a lack of transparency, and the possibility of data tampering. Because of these constraints, more dependable and secure solutions are required to guarantee confidence in digital systems.

Blockchain technology has emerged as a promising solution to address these challenges. Blockchain is a decentralized distributed ledger technology that records transactions securely across multiple nodes in a network. Once data is recorded in a block and added to the chain, it becomes extremely difficult to modify or delete due to cryptographic verification and consensus mechanisms [2]. This feature provides transparency, immutability, and improved security in digital environments. The concept of blockchain was first introduced through the Bitcoin system, which demonstrated how peer-to-peer transactions could be recorded without relying on a centralized authority [1]

The potential of blockchain technology to increase openness and trust in digital systems has drawn attention from all across the world in recent years. Blockchain applications for safe data exchange, financial transactions, and digital governance services are being investigated by numerous industries. By guaranteeing that data stored in the ledger cannot be changed without network consensus, research has demonstrated that blockchain can greatly improve system stability [3]. Blockchain has emerged as a crucial infrastructure for creating safe and open digital platforms as more businesses use decentralized technology.

This research aligns with the United Nations Sustainable Development Goals (SDGs), particularly SDG 9: Industry, Innovation and Infrastructure, which encourages the adoption of innovative technologies to build resilient digital infrastructures. In addition, the study contributes to the Distributed Systems and Blockchain Technology research cluster, which focuses on developing secure decentralized systems for modern digital environments.

1.2 Literature Review

Blockchain technology has been extensively studied in recent years as a secure infrastructure for distributed systems. Zheng et al. (2017) presented a comprehensive overview of blockchain architecture, describing key components such as distributed ledgers, consensus mechanisms, and

cryptographic security models [2]. Their study highlighted how blockchain enables transparent and tamper-resistant data storage across multiple network participants. Similarly, Casino et al. (2019) conducted a systematic literature review on blockchain applications and found that blockchain has been widely adopted in sectors such as finance, healthcare, supply chain management, and digital governance [3].

Yli-Huumo et al. (2016) looked more closely at the state of blockchain research and highlighted how it may increase transparency and lessen the need for reliable middlemen in digital systems [4]. Furthermore, Crosby et al. (2016) talked about how blockchain technology may support safe digital infrastructures in a variety of businesses and goes beyond cryptocurrency applications [10]. Additionally, platforms like Ethereum and Hyperledger Fabric have been created to facilitate enterprise-level blockchain systems and decentralized applications [6], [5].

Despite the benefits of blockchain technology, a few studies have shown flaws in current blockchain systems. Researchers and developers continue to have serious concerns about issues including scalability, high processing requirements, and integration difficulties with conventional systems [5]. These drawbacks emphasize the need for more investigation into blockchain architectures that can efficiently handle contemporary digital platforms while preserving performance, security, and transparency.

1.3 Research Gap

While the concept of blockchain technology has been extensively researched and implemented in various fields including finance, supply chain management, and digital governance, it has also been seen that most of the previously implemented blockchain technology solutions have been focused on individual applications rather than integrated digital systems. Most of the previously implemented blockchain technology solutions have also been seen to be operating as standalone solutions and have faced challenges while interacting with various technologies within the system. There have also been various research findings that have indicated the scalability issues and the need for higher computation while implementing the blockchain technology solutions.

Moreover, some existing blockchain systems such as Ethereum and Hyperledger offer a decentralized infrastructure, but these systems are technically challenging and require customization to integrate them well within a system environment. This poses difficulties in designing an efficient architecture that incorporates blockchain technology in a manner that is conducive to digital systems. Hence, there is a need to conduct research on how blockchain technology can be integrated to ensure a secure and transparent infrastructure within a comprehensive system architecture. The proposed approach is to incorporate blockchain technology within the system to ensure data integrity, transparency, and trust while providing ease of interaction with other components within the system.

Technologies and method	Research Paper [1] Bitcoin Blockchain	Research Paper [2] Bitcoin Blockchain	Research Paper [3] Bitcoin Blockchain	Research Paper [4] Bitcoin Blockchain	Research Paper [5] Bitcoin Blockchain	Proposed System
Blockchain base transaction recordings	✅	✅	✅	✅	✅	✅
Distributed ledger data storage	✅	✅	✅	✅	✅	✅
Decentralized Data verification	✅	✅	✅	✅	✅	✅
Enterprise Blockchain Architecture	❌	❌	❌	❌	✅	✅
Integration of Multi-Component system	❌	❌	❌	❌	❌	✅
Secure and Tamper resistance data Storage	✅	✅	✅	✅	✅	✅
Scalable Blockchain Infrastructure	❌	✅	❌	❌	✅	✅

Table 1 : Comparison of the existing research methods and proposed method

2.OBJECTIVE

2.1 Main Objective

The primary aim of this research is to develop a blockchain-based infrastructure that improves the security, transparency, and reliability of the entire system through the use of decentralized ledger technology. The use of a blockchain in the system will ensure that the data in the system is stored in a manner that is tamper-proof and verifiable.

2.2 Specific Objectives

- To study and analyze existing blockchain technology to identify the most suitable framework to be used to implement the blockchain component in the proposed system. This means evaluating existing blockchains such as Ethereum or Hyperledger in terms of security and compatibility.
- Design a blockchain-based architecture for the system, defining how data will be recorded, validated, and stored in a decentralized environment to ensure data integrity and transparency.
- To develop and implement the blockchain component using an appropriate blockchain platform, ensuring that the system can securely record and manage transaction data within the distributed ledger.
- To integrate the blockchain module with other components of the overall system, enabling secure communication and data exchange between the blockchain infrastructure and other modules in the platform.
- To evaluate the performance and security of the blockchain implementation, assessing factors such as system efficiency, scalability, and the ability of the blockchain component to maintain reliable and tamper-resistant records

3.METHODOLOGY

Design Science Research (DSR), which focuses on creating technology objects to address real-world issues, is the methodology used in this study. The goal of this research is to create and deploy a blockchain-based element that can improve the system's overall security, dependability, and transparency. Using this method, the research will entail determining the system requirements, creating the blockchain architecture, putting the solution into practice on appropriate platforms, and assessing its efficacy. Because design science enables the creation and evaluation of a technology solution inside an actual system environment, it is suitable for this research.

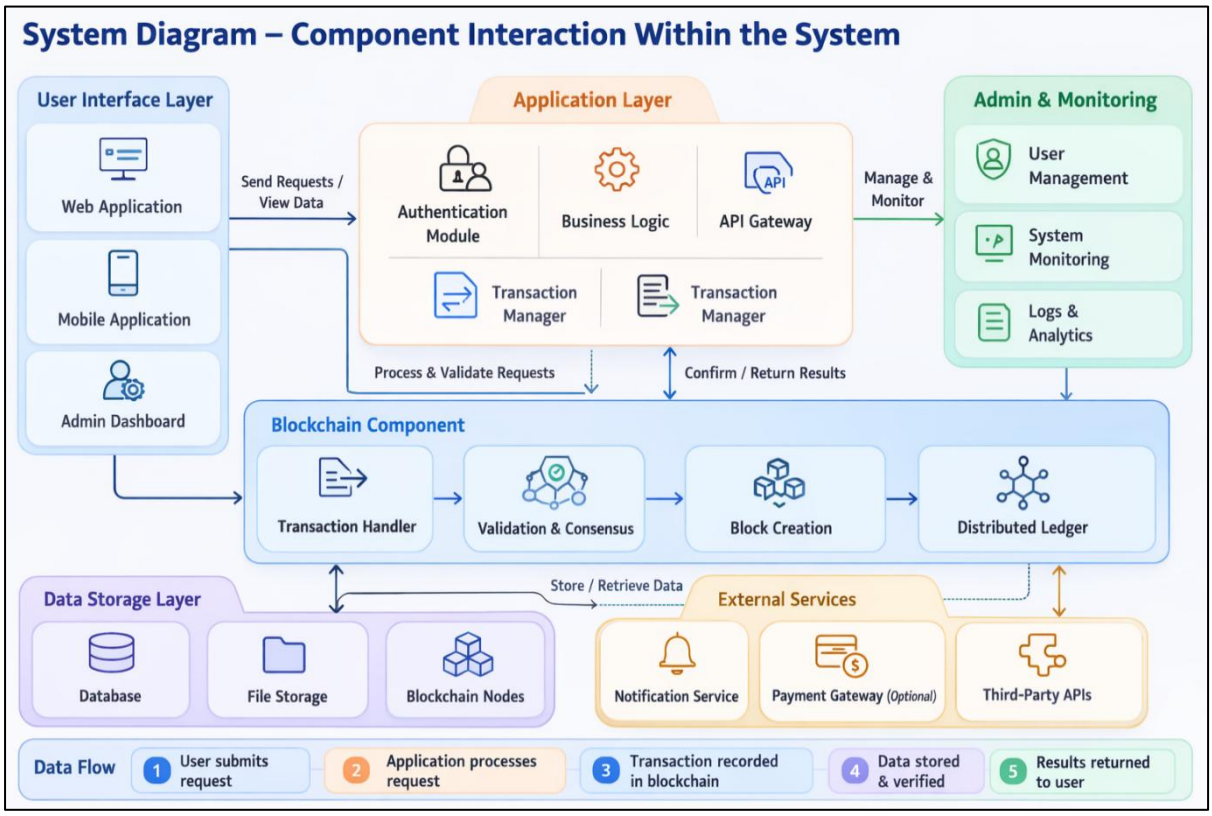


Figure 1 : System diagram-component interaction within the system

To safely preserve system transactions and data, the suggested system will make use of blockchain technology as a decentralized distributed ledger. Cryptographic hashing techniques will be used to store each transaction in a block and link it to earlier blocks. The integrity and immutability of stored data will be guaranteed by algorithms like SHA-256 hashing. Additionally, before transactions are posted to the blockchain network, a consensus process will be utilized to confirm them, guaranteeing that all users keep a reliable and consistent record of system activity.

Platforms like Ethereum or Hyperledger Fabric will be taken into consideration for the blockchain component's deployment. While Hyperledger Fabric offers permissioned blockchain architecture appropriate for enterprise operations, Ethereum offers a flexible environment for creating decentralized

applications. The blockchain network can be tested and deployed using development tools like Truffle, Ganache, and MetaMask. The blockchain module will be developed and integrated with other parts of the system using programming languages like Solidity, JavaScript, or Python.

To assess the blockchain infrastructure's performance, simulated or system-generated transaction data will be the main source of data used for system testing. During the research process, no private or sensitive information will be gathered. The responsible and secure handling of all test data shall uphold ethical principles. To avoid unwanted access or information misuse, the research will adhere to the proper data privacy and security procedures.

To validate the effectiveness of the blockchain implementation, several performance evaluation strategies will be applied. Metrics such as transaction processing time, system throughput, scalability, and data integrity verification will be measured to assess system performance. Security testing will also be conducted to ensure that the blockchain network successfully prevents unauthorized data modification. These evaluations will help determine whether the proposed blockchain component improves transparency, reliability, and security within the overall system.

4.HIGH-LEVEL SYSTEM ARCHITECTUR

In order to guarantee safe, transparent, and effective system operations, the suggested system is built as a multi-component digital platform that combines blockchain technology with additional auxiliary modules. The User Interface Layer, Application Processing Layer, Blockchain Component, and Data Management Layer are some of the major parts of the overall architecture. Users can monitor system outputs, submit requests, and interact with the system via the user interface. Communication between modules is coordinated and system logic is managed by the application processing layer. The data management layer houses system-related data needed for operational procedures, while the blockchain component serves as a decentralized ledger in charge of safely documenting system transactions.

The integrity and openness of system operations are crucially dependent on the blockchain component, which is the subject of this study. It is in charge of keeping a distributed ledger across participating nodes, recording transactions in blocks, and verifying them via the blockchain network. The transaction handler, block creation process, cryptographic hashing module, and distributed ledger storage are some of the internal components that make up the blockchain component. Together, these components guarantee that a transaction is immutable and verifiable throughout the network once it is recorded.

Below is a simplified **high-level system architecture diagram** that illustrates the interaction between components:

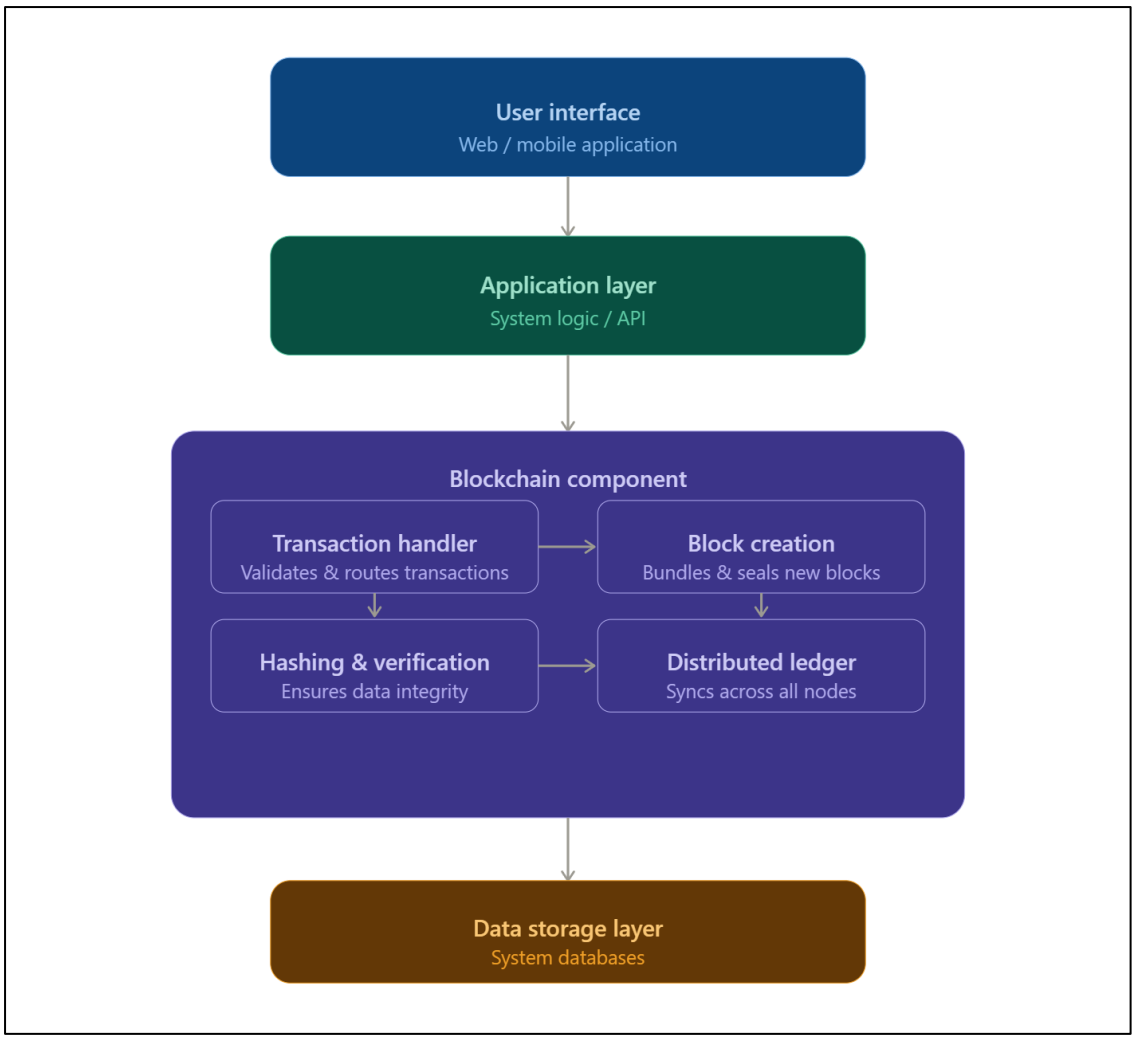


Figure 2 : blockchain system architecture diagram

When a user submits a request via the user interface, the components start interacting with one another. The request is processed and transaction data is prepared by the application layer. After that, the blockchain component receives this data, validates it, and records it in a new block. The block is dispersed over the blockchain network and connected to earlier blocks using cryptographic hashing. The transaction is permanently entered into the distributed ledger when it has been verified. After that, the application layer can obtain the pertinent data and use the interface to show the user the outcomes.

Scalability and security are taken into account when designing the system architecture. Blockchain technology offers a decentralized environment that boosts system resilience and lessens need on centralized servers. The distributed ledger improves system security and trust by guaranteeing that data cannot be changed without network consensus. Furthermore, the system's modular design makes it possible to incorporate additional services or components in the future without substantially altering the current architecture. As the system expands and adjusts to growing operational demands, this strategy promotes scalability.

5.USER REQUIREMENTS

Stakeholder identification

The proposed system involves several stakeholders who interact with the platform in different ways. The primary stakeholders include system users, system administrators, developers, and organizational decision makers.

- System Users – Individuals who interact with the system through the user interface to submit requests and access information recorded within the platform.
- System Administrators – Responsible for managing the system environment, monitoring system operations, and maintaining system performance and security.
- Developers / Technical Team – Responsible for designing, implementing, and maintaining the blockchain infrastructure and system modules.
- Organizations / Institutions – Entities that utilize the system for secure data management, transparency, and reliable transaction recording.

These stakeholders contribute to the successful operation and maintenance of the system.

Functional requirements

The functional requirements describe the core operations that the system must perform.

1. The system shall provide a user interface that allows users to interact with the platform and submit system requests.
2. The system shall process user requests through the application layer and prepare the transaction data for recording.
3. The system shall record transactions in the blockchain component to ensure transparency and immutability of data.
4. The system shall validate transactions using cryptographic verification methods before storing them in the blockchain ledger.
5. The system shall store and retrieve system data securely through the integrated data storage layer.
6. The system shall allow administrators to monitor system activities and manage system operations.

Non-functional requirements

Non-functional requirements describe system quality attributes such as security, performance, and usability.

Security

- The system shall ensure secure data storage using blockchain cryptographic mechanisms.
- The system shall prevent unauthorized access through authentication and authorization mechanisms.

Performance

- The system shall process transactions efficiently and ensure reasonable transaction confirmation time.
- The architecture shall support multiple user requests without system failure.

Usability

- The system shall provide a simple and user-friendly interface to allow easy interaction with the platform.
- Users should be able to access system features without requiring advanced technical knowledge.

Scalability

- The system architecture shall support future expansion and integration with additional modules.

Reliability

- The system shall maintain consistent data integrity and availability during system operations.



Figure 3 : User requirement interaction diagram

6.COMMERCIALIZATION PLAN

Target market and customer persona

The proposed blockchain-based system is primarily targeted at organizations and institutions that require secure, transparent, and reliable digital transaction management. Financial institutions, governmental bodies, private businesses, and digital service providers that handle substantial amounts of sensitive data and transactions are examples of potential users. Technology-driven businesses looking for increased security, trust, and transparency in their digital operations make up the primary customer profile. These clients frequently deal with issues like the possibility of data manipulation, a lack of transparency, and weaknesses in centralized systems. These companies can improve operational efficiency and system dependability by implementing blockchain technology.

Value proposition

The suggested approach offers a decentralized, safe digital infrastructure that enhances data integrity, transparency, and trust. The blockchain-based method guarantees that every transaction is documented in a distributed ledger that is impervious to manipulation, in contrast to conventional centralized systems.

The system's primary benefits include:

- Enhanced data security through cryptographic verification
- Improved transparency through decentralized record keeping
- Reduced risk of data manipulation and fraud
- Greater system reliability due to distributed network architecture

These features provide organizations with a more trustworthy digital environment for managing system operations.

Revenue model

The commercialization strategy may follow a Software-as-a-Service (SaaS) model, where organizations can subscribe to the platform to access blockchain-based services. Institutions can integrate the system into their existing infrastructure while paying subscription or service fees for system usage.

Additional revenue streams may include:

- Enterprise integration services for organizations requiring customized implementations.
- Maintenance and support services for system upgrades and security monitoring.
- API-based service access allowing external platforms to connect with the blockchain system.

Cost estimation

The estimated costs for developing and deploying the blockchain system include several components. These include system development costs, infrastructure costs, and operational maintenance costs. Development costs include software development tools, blockchain framework implementation, and testing environments. Infrastructure costs involve cloud hosting services and blockchain node management. Operational costs include system maintenance, security monitoring, and future system upgrades.

Initial development costs are relatively higher due to system design and implementation, but long-term operational costs are reduced because blockchain systems minimize the need for centralized intermediaries.

Pricing strategy

The pricing strategy will be designed to ensure affordability while maintaining sustainability for the system provider. The system may adopt a tiered subscription pricing model based on system usage and organizational size.

Possible pricing tiers may include:

- Basic plan – Suitable for small organizations with limited system usage.
- Standard plan – Designed for medium-scale organizations requiring higher transaction capacity.
- Enterprise plan – For large institutions requiring customized blockchain solutions and dedicated support.

This flexible pricing structure allows organizations to choose a service level that matches their operational needs

Competitive advantage

The proposed blockchain-based system offers several competitive advantages compared to traditional centralized systems. The key advantages include enhanced transparency, improved data security, and tamper-resistant record management.

The decentralized architecture reduces dependency on centralized authorities while increasing trust between system participants. Additionally, the system design allows easy integration with other digital platforms, making it adaptable for multiple industries and organizational environments.

Intellectual property considerations

The intellectual property related to the system includes the system architecture design, blockchain integration framework, and developed software modules. The developed solution may be protected

through software copyrights, system design documentation, and potential patent applications if innovative methods are introduced.

Proper intellectual property management ensures that the creators maintain ownership of the developed technology while allowing controlled licensing or commercial deployment of the system.

7.BUDGET AND JUSTIFICATION

The development of the proposed Blockchain-Integrated Microfinance & Welfare Management System requires resources such as internet connectivity, electricity, cloud infrastructure, and development tools. The system will use blockchain technology and microservices architecture, which requires cloud services for hosting containers and backend services. The estimated budget required for the research project is shown in Table 4.

Resource	**Cost (LKR)**
Internet	4,000
Electricity	5,000
Cloud Services (AWS / Azure for microservices & blockchain testing	8,000
Software Tools & Development Resources	5,000
Transport	6,000
Communication	2,000
Research Paper Publication	5,000
Total Estimated Budget	**35,000**

Table 2 : Table of budget

8.WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) divides the project into smaller tasks assigned to each group member. This structure helps ensure balanced workload distribution and proper coordination among team members. Each component contributes to the development of the overall system, and tasks are scheduled according to the project timeline.

Project Components

- Blockchain-Based Distributed Data Layer
- Smart Contract & Eligibility Automation
- AI-Based Microfinance Prediction Engine
- AI-Based Microfinance Prediction Engine

Member Component	Tasks
Blockchain-Based Distributed Data Layer	Study blockchain technology, design blockchain architecture, implement blockchain network (Hyperledger Fabric), develop on-chain and off-chain data storage, test transaction recording and data integrity
Smart Contract & Eligibility Automation	Design smart contract logic, implement smart contracts, automate rule-based verification process, test smart contract execution
AI-Based Microfinance Prediction Engine	Collect datasets, perform data preprocessing, develop machine learning model, implement risk prediction systems, evaluate model performance
NLP-Powered Multilingual Communication System	Design NLP architecture, develop chatbot for multilingual communication (Sinhala, Tamil, English), implement user query processing, integrate with system modules

Table 3 : Task Breakdown Per Member

Phase	Activities
Phase 1	Literature review and requirement analysis
Phase 2	System architecture design
Phase 3	Development of individual components
Phase 4	System integration and testing
Phase 5	Performance evaluation and documentation

Table 4 : Timeline Alignment

The project workload is evenly distributed among the four group members. Each member is responsible for developing and implementing a specific component while collaborating during system integration and testing phases. Regular meetings and progress evaluations will ensure that all tasks are completed according to the project timeline and that the system components function together effectively.

9.GANTT CHART

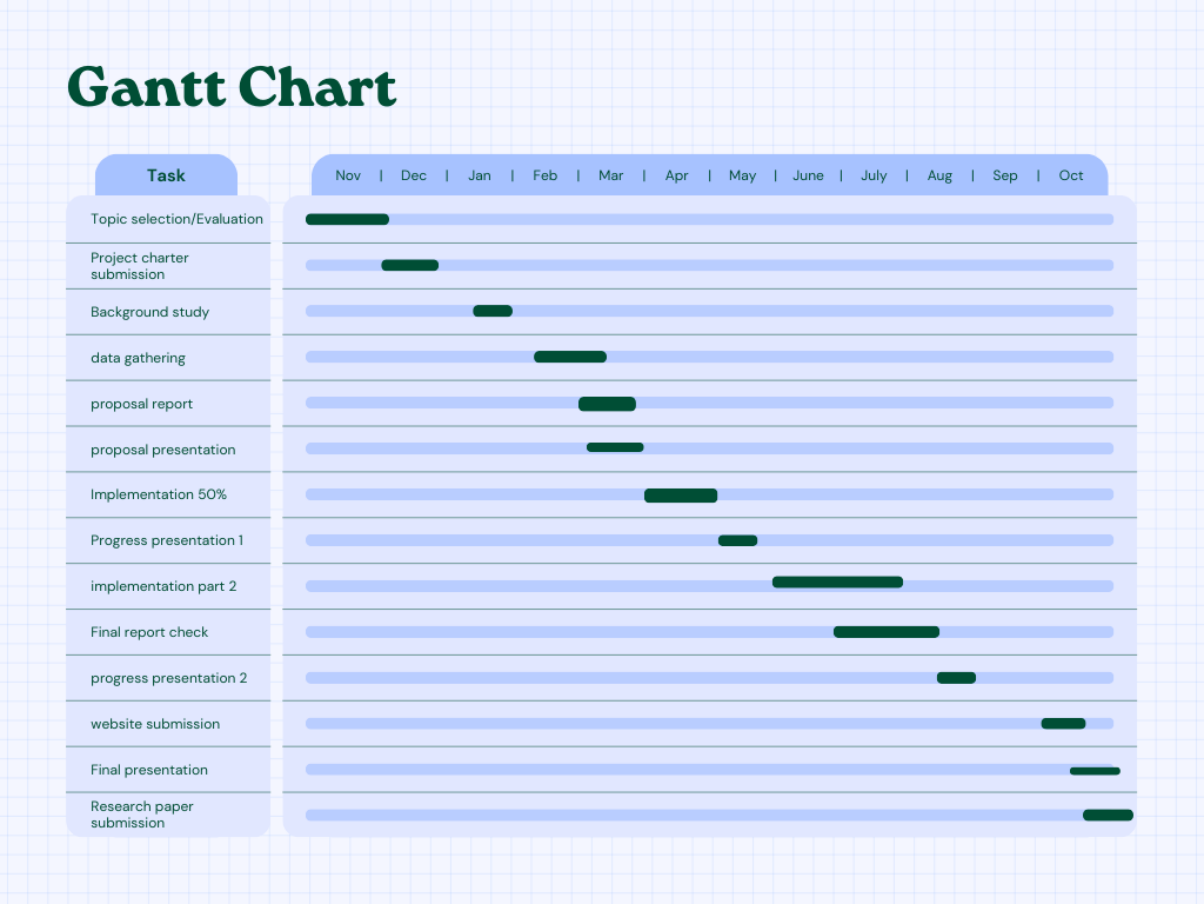


Figure 4 : Gantt Chart

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10.APPENDICES

Appendix A

Loan Application Data Attributes (Provided by Samupakara Gramiya Sanwardhana Bank)

Sample Ledger Portfolio (Microfinance Credit Scoring)

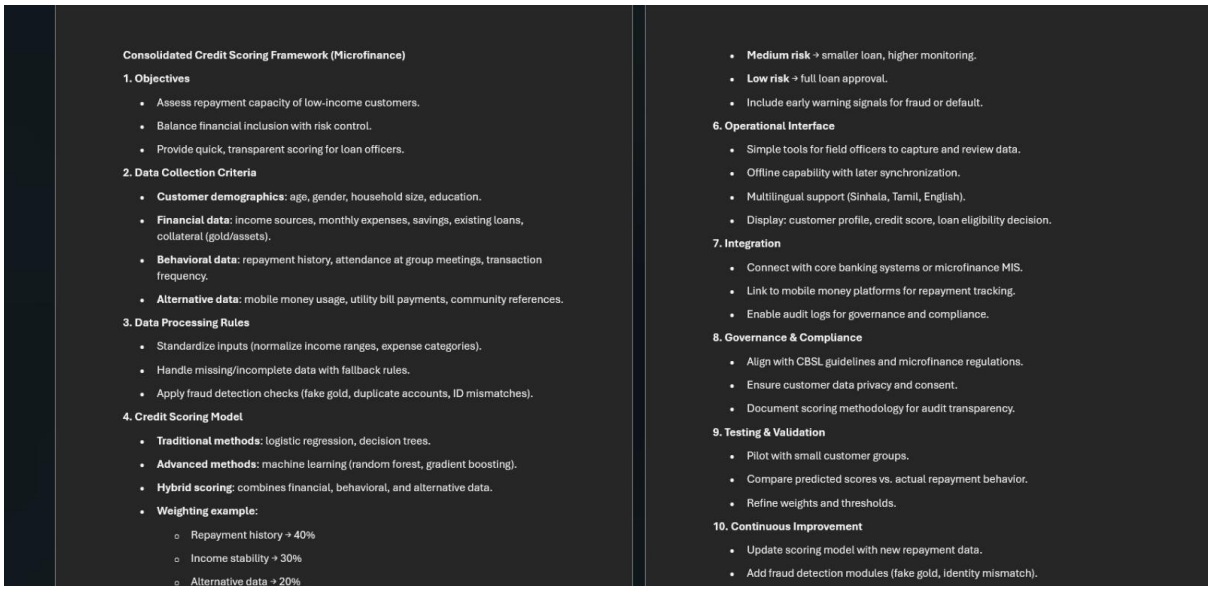
Field	Details / Example	Purpose
Customer ID / Account No.	2025-MF-001	Unique identifier
Customer Name	John Silva	Identification
National ID / Passport No.	123456789V	Verification
Age / Gender	35 / Male	Demographics
Address / Location	Colombo, Sri Lanka	Regional mapping
Household Size	4	Dependency ratio
Loan Type	Gold-backed / Micro loan	Product category
Loan Amount	LKR 50,000	Exposure
Collateral Details	22K Gold chain, 25g	Security
Monthly Income	LKR 35,000	Repayment capacity
Monthly Expenses	LKR 20,000	Debt-to-income ratio
Savings / Deposits	LKR 5,000	Financial discipline
Existing Loans	LKR 15,000 (other MFI)	Over-indebtedness check
Repayment History	95% on-time	Reliability
Attendance (Group Meetings)	90%	Commitment
Mobile Money Usage	Active (weekly transactions)	Alternative data
Utility Bill Payments	On-time	Responsibility
Risk Flags	None	Fraud detection
Credit Score	78/100 (Band B)	Risk rating
Decision	Approved – Conditional	Loan eligibility
Officer Remarks	Customer reliable, collateral verified	Governance
Date of Entry	08 Dec 2025	Audit trail

✅ **Key Notes**

- This **ledger portfolio** can be digitized into your app database.
- Each row = one customer record.
- You can add **columns for audit logs** (who entered, who approved).

Appendix B

Bank Requirement Document



Consolidated Credit Scoring Framework (Microfinance)

1. Objectives

- Assess repayment capacity of low-income customers.
- Balance financial inclusion with risk control.
- Provide quick, transparent scoring for loan officers.

2. Data Collection Criteria

- **Customer demographics**: age, gender, household size, education.
- **Financial data**: income sources, monthly expenses, savings, existing loans, collateral (gold/assets).
- **Behavioral data**: repayment history, attendance at group meetings, transaction frequency.
- **Alternative data**: mobile money usage, utility bill payments, community references.

3. Data Processing Rules

- Standardize inputs (normalize income ranges, expense categories).
- Handle missing/incomplete data with fallback rules.
- Apply fraud detection checks (fake gold, duplicate accounts, ID mismatches).

4. Credit Scoring Model

- **Traditional methods**: logistic regression, decision trees.
- **Advanced methods**: machine learning (random forest, gradient boosting).
- **Hybrid scoring**: combines financial, behavioral, and alternative data.
- **Weighting example**:
  - Repayment history → 40%
  - Income stability → 30%
  - Alternative data → 20%
- **Medium risk** → smaller loan, higher monitoring.
- **Low risk** → full loan approval.
- Include early warning signals for fraud or default.

6. Operational Interface

- Simple tools for field officers to capture and review data.
- Offline capability with later synchronization.
- Multilingual support (Sinhala, Tamil, English).
- Display: customer profile, credit score, loan eligibility decision.

7. Integration

- Connect with core banking systems or microfinance MIS.
- Link to mobile money platforms for repayment tracking.
- Enable audit logs for governance and compliance.

8. Governance & Compliance

- Align with CBSL guidelines and microfinance regulations.
- Ensure customer data privacy and consent.
- Document scoring methodology for audit transparency.

9. Testing & Validation

- Pilot with small customer groups.
- Compare predicted scores vs. actual repayment behavior.
- Refine weights and thresholds.

10. Continuous Improvement

- Update scoring model with new repayment data.
- Add fraud detection modules (fake gold, identity mismatch).